L’SPACE Mission Concept

Preliminary Design Review Document

Descent, Lander, and Payload Experiment Criteria

The PDR demonstrates that the overall program preliminary design meets all requirements with acceptable risk and within the cost and schedule constraints and establishes the basis for proceeding with detailed design. It shows that the correct design options have been selected, interfaces have been identified, and verification methods have been described. Full baseline cost and schedules, as well as all risk assessment, management systems, and metrics are presented. (NPR 7120.5D p.30)

The PDR document is required to follow the order of sections as they appear below.

Preliminary Design Review Report

I) Summary of PDR report

1.1 Team Summary

Aaron Gonzalez

- Project Manager
- Texas State University
- Skills
    - CAD, Prototyping
- Majoring in Physics

Jason Cabrejos

- Deputy Project Manager, Science Team Member
- University of Texas at Arlington
- Mechanical Engineering
- Skills: Solidworks, MATLAB, Microsoft Office

Yadiel Santos

- Administration Lead, Science Team Member
- Texas State University
- Physics, Secondary Education
- Skills: Microsoft Office

Omar Asbun

- Safety Officer, Engineering Team member
- University of Texas at Arlington
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- Skills: MATLAB, Solidworks, Excel, Word

Lauren Savage

- Engineering Team Lead

- University of Texas at Arlington
- Aerospace Engineering; Mechanical Engineering (minor)
- Skills: MATLAB, Python, SolidWorks, ANSYS FEA & CFD, Microsoft Office Suite, Systems Engineering Modeling & Requirements

Daniel Haque

- Science Team Lead
- University of Texas at Austin
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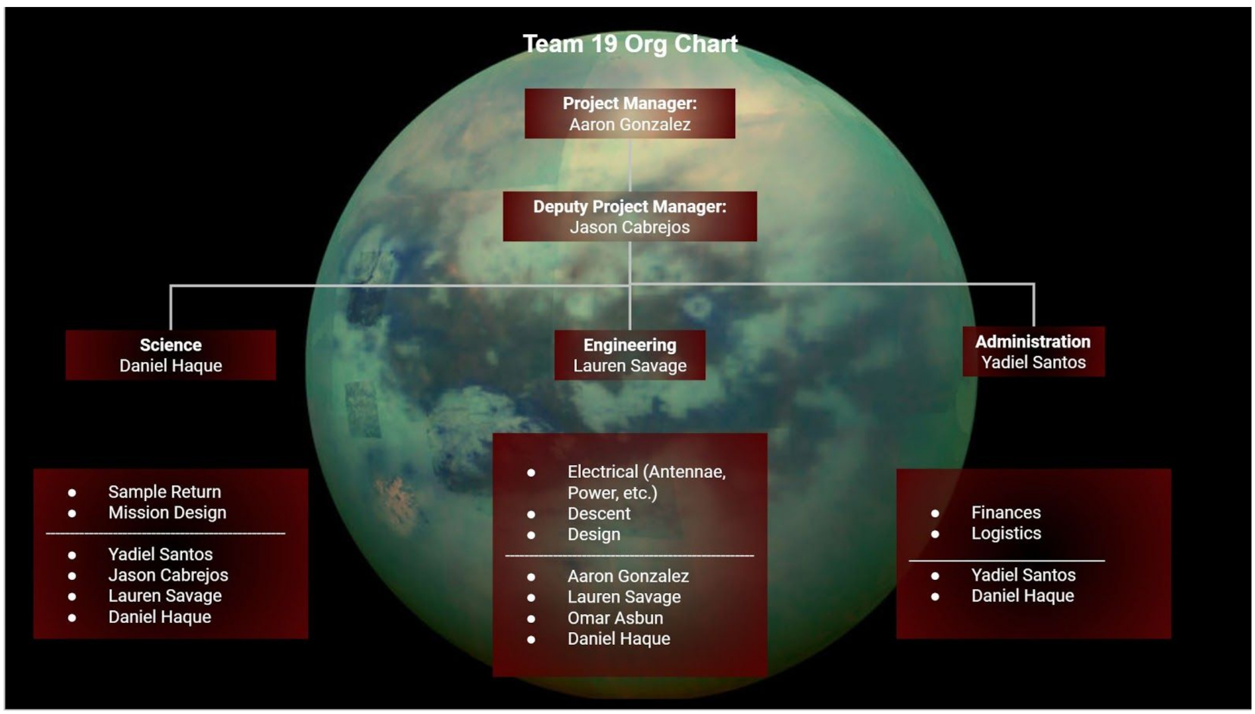
- Skills: Python, MATLAB, Soldering, Microsoft Office



1.2 Descent and Lander Summary

The descent and lander design investigated and developed in this mission concept resembles the engineering descent system used in NASA's previous space shuttle missions. The vehicle will enter Titan's atmosphere on an interplanetary transfer from Earth, into Saturn's sphere of influence, and from a 100 km parking orbit around Titan. Once positioned in the parking orbit, the spacecraft will initiate the entry sequence by firing the reaction control system (RCS) thrusters to place the vehicle into position for descent. The de-orbit burn will fire the rockets in the orbital maneuvering system against the vehicle's direction of motion in order to slow the vehicle enough so that it will be overcome by the force of gravity and pulled towards Titan. The RCS thrusters will rotate the spacecraft 180 degrees clockwise so that the vehicle enters the atmosphere “belly-down” at an attitude of 40 degrees. Making a series of “S” turns to starve off energy from the descent, the vehicle will approach the desired landing location on Kraken Mare, and prepare for a splashdown. The lander will deploy a parachute to slow the vehicle after starving off hypersonic reentry speeds to slow the vehicle to an appropriate landing speed, prior to splashing down in Kraken Mare (70°N, 310°W). Following splashdown, the fairing will break open into two pieces and allow the submarine to float autonomously in the lake.

The submarine diameter is 1963.44 mm across and 1327.48 mm tall. In order to accommodate the submarine this size, the landing system will house the submarine sitting upright. The size of the landing system was designed to house the submarine, with a margin of leeway. The landing system's cargo was sized to be 1950 mm in width and a length of proportionate size.

1.3 Payload Summary

Priority	Task	Heritage	Instruments
1	Sample Liquid Methane-Ethane	IR Spec - Spitzer Space Telescope	IR Spectrometer (IR Spec) Force Detected Nuclear Magnetic Resonance (FDNMR)
2	Find life	SC - Mastcam-Z on NASA's Mars Curiosity rover. IR Spec - Spitzer Space Telescope	N-type Thermocouple Temperature Sensor Simple Camera (SC) Infrared spectroscopy (IR Spec)
3	Map Mare Floor	SC - Mastcam-Z on NASA's Mars Curiosity rover. IR Spec - Spitzer Space Telescope	Side Scan Sonar (SS) Simple Camera (SC)

4	Bed Composition	APXS - on board both the Mars Exploration Rovers that arrived on the red planet in January, 2004	Alpha Particle X-ray Spectrometer (APXS) Benthic Sample Acquisition (BSA) Infrared spectroscopy (IR Spec)

Experiment 1: Main concern of this mission, to examine the liquid composition on kraken mare. The Force-Detected Nuclear Magnetic Resonance (FDNMR) and the IR Spectrometer (IR Spec).

Experiment 2: There has been speculation over the viability of life in Titan's mares, the FDNMR, SC, IR Spec, and Temperature Sensor are employed to explore this concept

Experiment 3: No satellites or past missions have been able to successfully check how deep the mare floor is or its geological structure. With the Side Scan Sonar (SS) and Simple Camera (SC), the mare floor would be mapped

Experiment 4: Checking the bed composition of the mare floor is the most difficult, it relies on the performance of the Benthic Sample Acquisition (BSA), from there it passes the sample from either the Alpha Particle X-ray Spectrometer (APXS), or Infrared Spectroscopy (IR Spec).

II) Evolution of Project

2.1 Highlight all changes made since the initial concept was identified and the reason for those changes.

The beginning of the concept of the mission started off rocky because the mission concept was still not 100% clear. So the first idea that came up after doing some research was to create a drone that would fly through the thick atmosphere and collect data with onboard equipment. After attending more meetings and learning more about the mission and Titan, some plans for a submersible rover were drawn up. The plan was for the rover to either drop down near Kraken Mare or Ligeia and explore one of the lakes until the initial mission was complete. After the initial mission if another mission was funded then the craft would travel to the other lake while collecting data and continue experiments until the end of its next mission. Finally after performing more research and attending even more meetings, it was decided that the landing zone would be around or in Kraken Mare and the mission would be restricted to just that lake. The design for the rover was scrapped, but the design for the submarine was kept and has been built off of since then with a Kraken Mare landing at 70°N, 310°W.

Changes to Payload Criteria:

The current payload, located on section 1.3, took a week and a half of research.

Experiment 1: Needing to sample the liquid on Titan, a few options were available. The Gas Chromatography-Mass Spectrometer (GCMS) , Liquid Chromatography-Mass Spectrometer (LCMS), Tandem Mass Spectrometer (MS/MS), Alpha Particle X-ray Spectrometer (APXS). The GCMS is normally used to identify trace elements thought to have disintegrated beyond identification however it requires a carrier gas of 99.99% Helium and is thus not possible to take and maintain on titan. The APXS is used on the Mars Rovers to detect the composition of solid objects, it does not meet the goals of this experiment to determine the composition of liquids. The MS/MS would be too expensive, being as each MS costs about 50M. Taking about 28.6% of our budget as where traditionally only 1 MS is used on Mars Missions. The LCMS is normally used analyze biochemical, organic, and inorganic compounds commonly found in complex samples of environmental and biological origin. The instrument also requires a high purity of nitrogen gas, Titan boasts a nitrogen 95% which would be no problem to maintain in the event of a leak.

Experiment 2: Finding life in a liquid boasts its own problems in terms of automation. Instruments found were the, Sonar, Temperature Sensor, Microscope, Camera, Infrared Spectroscopy (IR Spec), and Tandem mass spectrometry (MS/MS). Using a technique called passive sonar, the sensor acts as a giant ear in which it can listen for life, however if there happens to be life on Titan it is theorized to be simple organisms which would produce almost no sound. Therefore the idea for passive sonar to locate life was abandoned but will be used in Experiment 3. The temperature sensor chosen is an N-type Thermocouple which has a range that down goes to -270C. It will be used to detect ‘hot spots’ similar to earth’s ocean volcanic life. The microscope posed its own challenge, specifically in sterilizing the slides, given the small sample size and difficulty to teach the AI how to search for biological organisms, the idea was abandoned. The Camera was going to go on the craft in one way or another, in terms of life it can confirm the location of ‘hot spots’ detected by the thermocouple. It is theorized that the visibility in the Methane-Ethane mare is near clear. Water is a universal solvent whereas Methane-Ethane struggles to dissolve many elements and materials, thus making the mares easier to see in. The IR Spec can be used to reveal the structures of covalently bonded chemical compounds through their absorption of different frequencies of the infrared wavelength and can be used with both liquid and solid samples. As mentioned before the MS/MS is too expensive for this mission.

Experiment 3: Mapping the Mare floor posed the options of Sonar, Depth Sounder, Hyper-Spectral Sensor, and Radar. Using a technique called active sonar, the submarine will let out a ‘ping’ and from the time it takes to reach the ocean floor and back, it can calculate distance. The specific type of sonar is called Sidescan Sonar (SS). The Depth Sounder works on the same principle as the sidescan sonar. The Hyper-Spectral sensor is normally used on satellites to study changes in the environment but tend to be expensive and complex and not normally used to map the submerged floors. Finally Radar works on a principle similar to Sonar but uses electromagnetic radiation in the microwave band. In fear of using microwave frequencies on potential living cells and violating Experiment 2, radar will not be used.

Experiment 4: Determining the composition of the mare bed is the most difficult to accomplish due to having to detect it through a liquid. Using a Benthic Sample Acquisition (BSA), composed of a linear actuator and a specialized gripper, the sample can easily be acquired and sent to multiple instruments. The instruments options were the Alpha Particle X-ray Spectrometer (APXS), Microscope, Radar, Infrared Spectrometer (IR Spec). The APXS uses alpha particles and is used on the mars missions to analyze rock and soil samples, it has a low cost, and is relatively small compared to other instruments. Microscope, as mentioned in Experiment 2, is too difficult to automate. The Ground Penetrating Radar as mentioned in Experiment 3, will not be used due to its use of electromagnetic waves in the microwave spectrum. The IR Spec, as mentioned before in Experiment 2, reveals the structures of covalently bonded chemical compounds through their absorption of different frequencies of the infrared wavelength, in this case it will be used for solid samples.

Changes to Mission Experiment and Implementation Plan:

With the solidification of the submarine idea, there were a number of experiment ideas. Such as:

A. Life in methane-ethane mares
B. Lake depth in ethane mares
C. Lake bed material
D. Lake material distribution liquids
E. Growing Lakes and their rings
F. Titanquakes
G. Origin of methane in the atmosphere
H. Magic island
I. Weather and how its affected by the sun and Saturn’s Magnetic Field
J. Existence of the global ocean

With the current technology in ground penetrating radar, most of which only have a range of up to about 10 feet, nowhere near far enough to confirm the existence of a global ocean canceling idea J. Idea I is difficult to observe from the surface and is more suitable for a satellite. Idea H could be plausible as an extended mission with the simple camera, but not a current priority. Idea G is also plausible as an extended mission but not a current priority, it was voted out during a group meeting. Idea F is difficult to obtain accurately within a lake and was also voted out during a group meeting. Idea E is plausible as an extended mission but was also voted out as a priority during a group meeting. Ideas A, B, C, D were voted in during a group meeting as being the most plausible and all instruments are specialized for it.

III) Descent and Lander Criteria

3.1 Selection, Design, and Verification of Mechanical Descent and Lander Mechanism

The mission of the descent and lander system is to transport the submarine safely from a parking orbit around Titan, through the atmosphere, and finally to a splashdown in Kraken Mare, where the submarine will be released from the landing capsule.

The design requirements for the lander depends on many factors, and must be determined with the overall Titan mission, including scientific instruments and payloads in mind. The requirements of the mission system are dependent on the following:

- Payload
- Flight rate
- Propulsion
- Scientific instrument configuration constraints
- Planetary/environmental considerations

The two aspects of the missions that the descent and lander will perform are considered: the first aspect considering the overall mission scenario within which the stage would be operated. Overall energy requirements, basing options for the stage, mission duration, and the type of mission operations are associated with the transportation of scientific payloads to the surface of Titan. The second mission aspect considered is the nature of the scientific payloads themselves.

The major milestone schedule consisted of project initiation, design, manufacturing, verification, operations, and major reviews. The manufacturing project of the lander and descent mechanism was initiated after a series of literature reviews on past NASA and ESA planetary and lunar design systems. Since Kraken Mare was chosen as the sole destination for this mission, there was no need for a rover-based delivery system or any land-based mobility--the sole transportation method on the surface of Titan is restricted to navigation through Kraken Mare. As there was no need for a rover or other land-based transportation systems, a splashdown was selected as the mission's landing criteria. The submarine was designed to be cryogenic and buoyant in the mostly ethane-based Kraken Mare--but it does not contain any thermal protection systems and would not survive entry from orbit; therefore, a lander is required to protect the cryogenic vessel and its payload.

The major milestone schedule contains the project initiation, design, manufacturing, verification, operations, and major reviews. The project initiation consisted of reviewing the mission requirements, and establishing the major scientific objectives. The landing site was selected to be Kraken Mare, and based on this site and the determination to test and survey the lakebed material as well as the lake biology, the vehicle was selected to be a submarine. A submarine was an ideal design to implement because the while the depth of Kraken Mare is limited to the images we have received from Cassini's survey of Titan, it is assumed to be at least 115 ft. deep. A submarine can survey these depths, the topology of the lake, and carry a sufficient load of scientific instruments needed to analyze the chemistry and any signs of life. The system's design was therefore based on the needs of a cryogenic submarine, which could withstand the hydrocarbon lake's plunging temperatures of -250 degrees F. In order to withstand these cryogenic temperatures, the hydrocarbon composition, and the pressure accumulated from descending in Kraken Mare, the primary submarine material was chosen to be 9% nickel steel. The descent system requires an adequate thermal protection system and can therefore be comprised of titanium, steel, and composite ceramic tiles. A descent system is therefore required for safe delivery of the cryogenic submarine to Kraken Mare. In order to safely deliver the submarine to Kraken Mare, the descent vehicle, stationed in a parking orbit around Titan will perform a series of positioning burns to align the vehicle for proper atmospheric entry. The vehicle will then begin a series of de-orbit burns to enter the atmosphere, flying in a series of "S" turns to starve off energy as the vehicle reaches the desired landing coordinates within Kraken Mare at 70 degrees North, 310 degrees West.

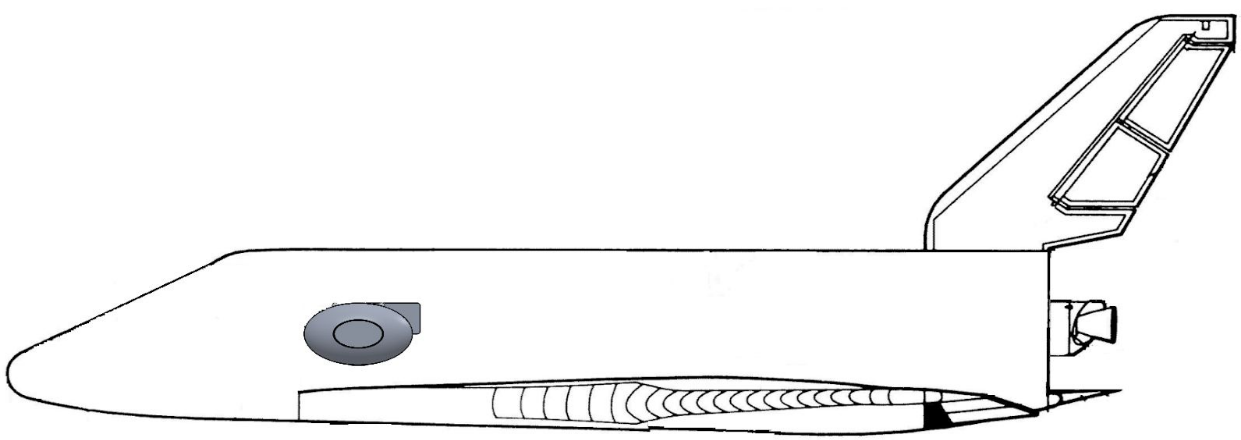
The propellor used on the cryogenic submarine is a COTS propellor, manufactured from steel. The performance of the materials' ability to withstand high temperature fluxes were evaluated during the testing phase of this program. The requirements for the lander system include the ability for the metal to allow expansion as well as adequate thermal protection during the orbital entry period. The space propulsion requirements include the needs for proper Reaction Control Thrusters to be fired at the appropriate time to rotate the craft 180 degrees into position and fire the de-orbit burns to initiate atmospheric re-entry.

3.2 Recovery Subsystem (Highlighted Because of Criticality)

The risks demonstrated in this mission are related to the possible malfunctions in the following systems:

- Propulsion (space & Kraken Mare)
- Thermal Protection Systems (landing system and submarine)
- Parachute Ejection

The purpose of establishing a testing schedule well before launch is to ensure the mission will be successful in all of its components. Testing done for the space propulsion components includes thrust and control verifications for the RCS thrusters. Whereas testing done on the COTS propellor included a series of evaluations using the same rudder in a high-density liquid, responsible for propelling great mass. The thermal protection systems for the lander system were tested by ensuring the ceramic composite tiles did not allow the spacecraft material beneath to strain and heat beyond a maximum requirement. This was accomplished using a high-temperature, high-pressure environment to replicate re-entry. The thermal protection system for the cryogenic submarine is simply the metal used for the exterior hull -- 9% nickel steel. The submarine was tested in a high-density, high-pressure, low-temperature testing environment to replicate the extreme environment found submerged under the surface of Kraken Mare. The landing delivery system adapted from a glider re-entry vehicle concept, containing RCS thrusters and a composite matrix is shown below. The submarine (payload) is shown in proportion to the size of the landing system.



Analysis was used to determine size for mass, attachment scheme, deployment process, and test results. In order to determine size for mass, the vehicle materials were taken into account, including electronics and scientific instruments. The entire lander and delivery system, including the submarine, were constrained to stay within weight restrictions--and minimizing the size of the systems mitigates the need for more materials.

The submarine was delivered to Titan attached within the landing system. Upon splashdown, the landing system was programmed to break open and gently release the submarine into the mare.

3.3 Mission Performance Predictions *(Highlighted Because of Criticality)*

The baseline for mission success consists of using the front camera on a gimbal to take a 150 degree panorama. Each photo on the camera takes approximately 0.25 seconds. To complete a panorama would take about 10 seconds to complete.

Below is a list of the instruments and their Optimal times. After the baseline is completed, experiments will be completed based off priority as stated in section 1.3.

Instrument	Optimal Time (time to collect data)
FDNMR	10 seconds to 10 min
APXS	15 min to 10+ Hours (30 minutes avg)
SS	0.21 sec
IR Spec	5-10 minutes
Simple Camera	0.25 sec (4 fps)

The minimum science and engineering requirements for NASA to consider this mission successful are the following:

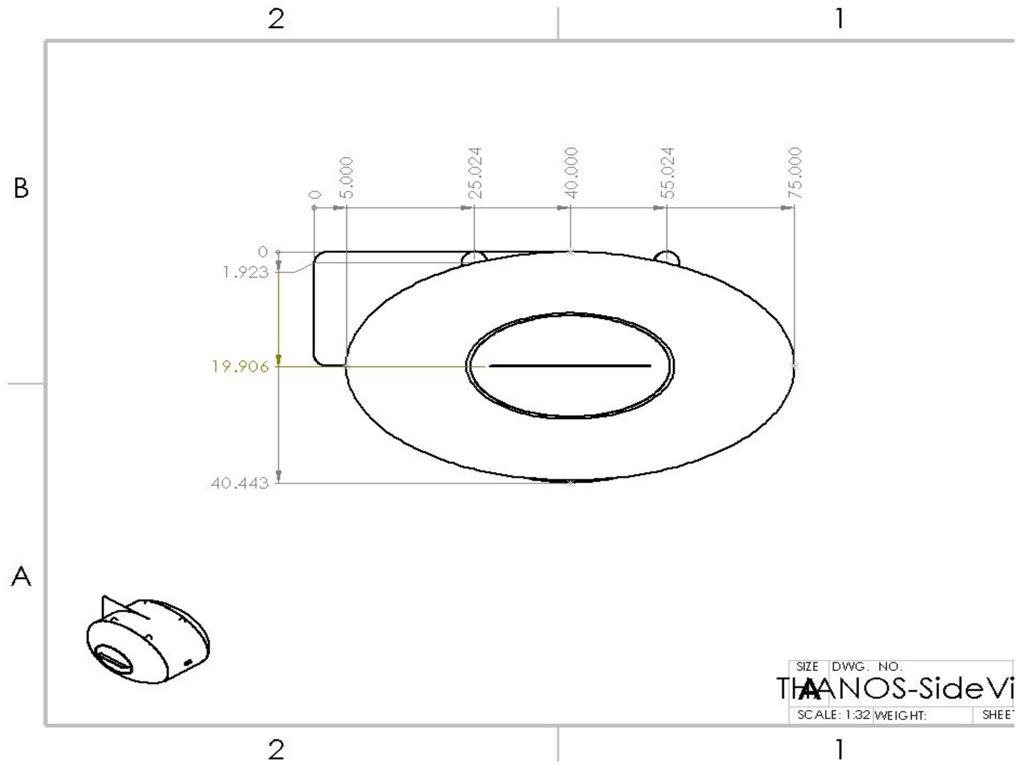
- The mission shall survey the depths of Kraken Mare and map the full topological

layout of the lakebed, including full depth measurements.
- The mission shall sample the lakebed in multiple locations in order to analyze the chemical and geological composition of the hydrocarbon lake and Titan's surface.
- The mission shall observe Kraken Mare for any signs of life.

Seasonal weather patterns on Titan are similar to Earth's seasonal patterns. There is a winter and a summer, and solstices in between. However, one major difference is that seasons on Titan last for 15 years each. The winter season on Titan is marked by larger hydrocarbon lakes in the northern hemisphere, decreased haze around the equinoxes, and ice clouds in the south polar regions. Dunes of soot-like material rains down from the atmosphere in the equatorial regions during the equinox. Violent storms producing downdrafts of up to 10 m/s along with methane storms have been observed to occur during the equinox. During the northern hemisphere summer, turbulence across the hydrocarbon lakes were observed, suggesting an increase in surface winds during the Titan summer. Therefore, based on the above observations, the ideal season for the entry into Titan is during the winter season. This season was chosen because the hydrocarbon lakes in the northern hemisphere are observed to be larger, equatorial haze tends to be reduced, and surface winds appear to be calm--including reduced chances for down-drafts and turbulent lake conditions. As this mission will be performed using a submarine, it is best to establish the mission before the equinox.

The hydrostatic pressure the submarine will need to withstand, assuming a maximum depth of 35.052 m is 30803.698 kg'ms^2. However, these assumptions are based on the imagery provided by Cassini, and are not confirmed. Additionally, the average depth needed to descend to would be around 15.24 m as the lakebed samples can be taken from more shallow areas of Kraken Mare. Therefore, the maximum hydrostatic pressure the submarine will likely endure while sailing through Kraken Mare is 13392.912 kg/ms^2.

3.4 Payload Integration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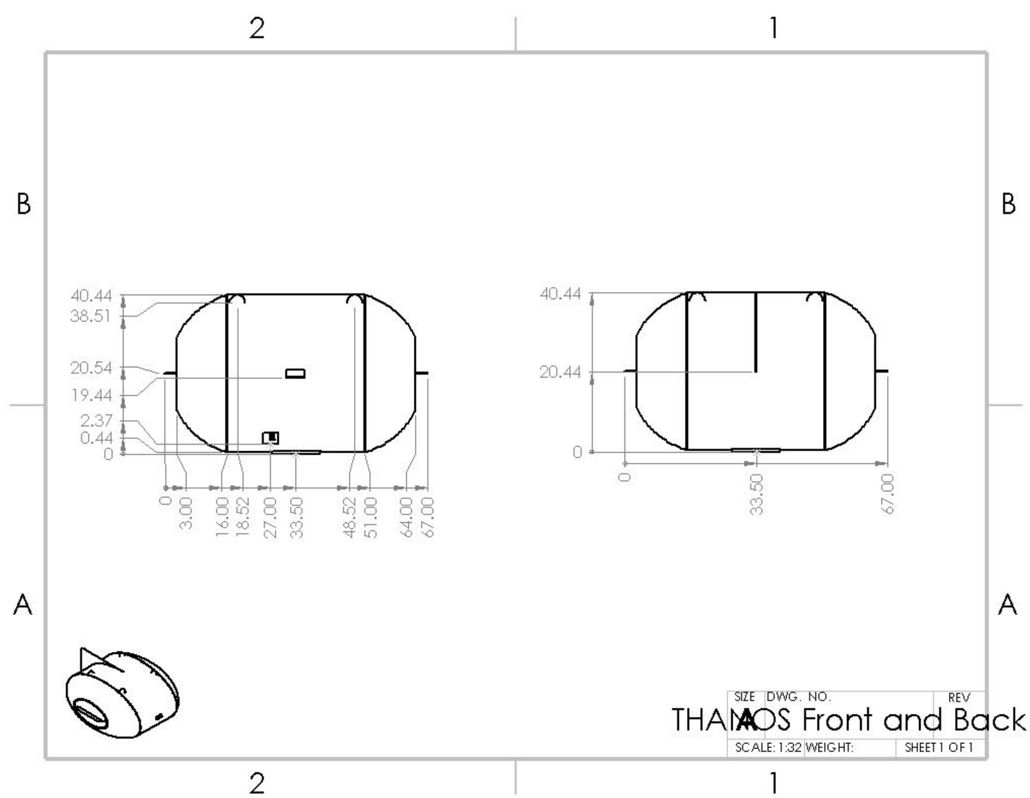
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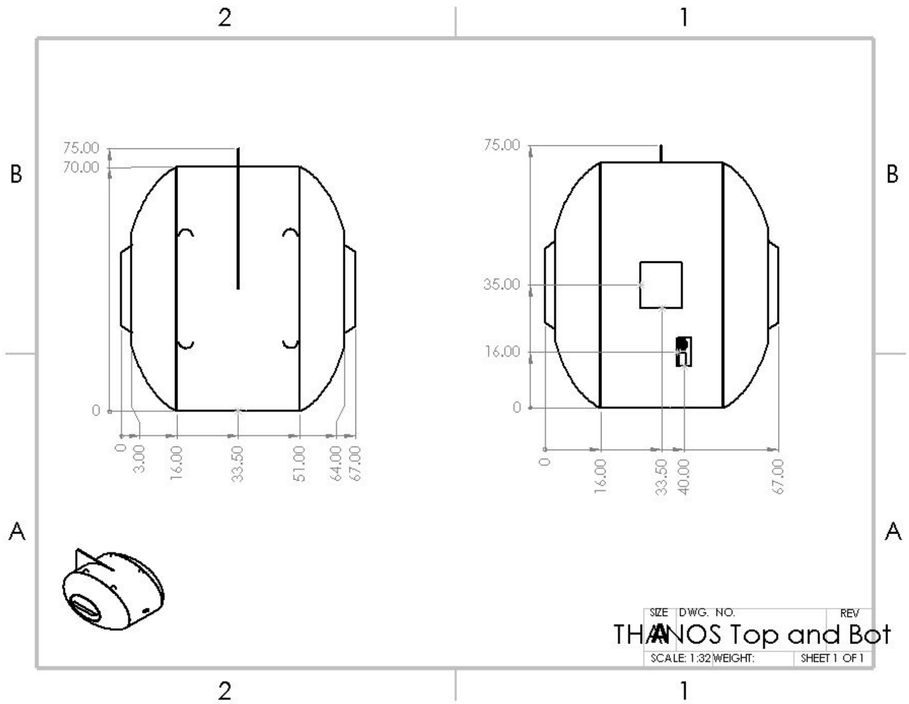
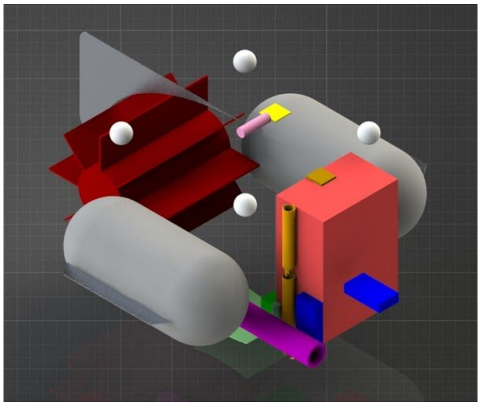
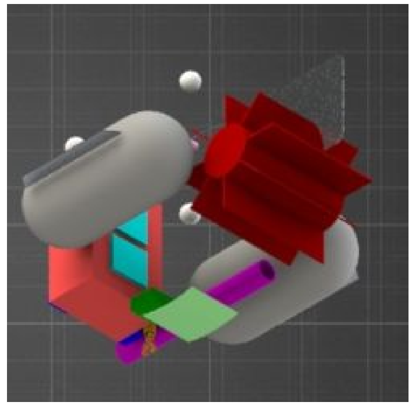
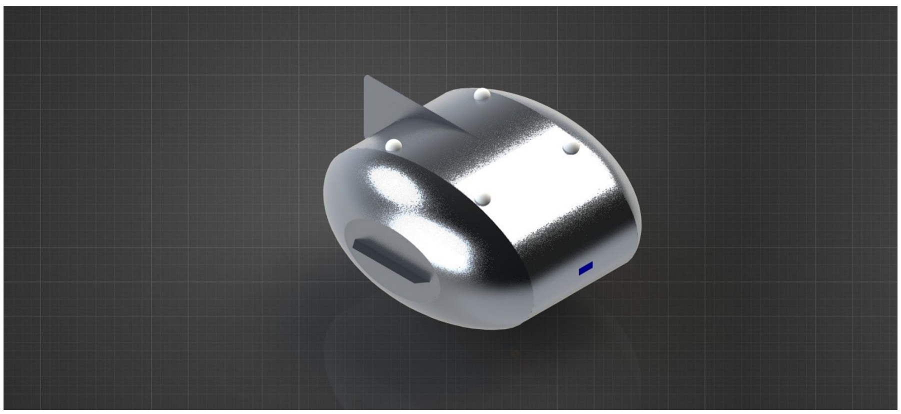


Dimensions are in inches

The images below are a rendering of THANOS and all the instruments included. The instruments are color coded as follows

- Ballast Tanks - Grey
- MMRTG - Dark Red
- RCE - Cyan
- Battery - Dark Green
- Low Gain Antenna - Pink
- UHF Antenna - Yellow
- Transceiver - Brown
- Simple Camera - Blue
- BSA - Orange
- FDNMR - Purple
- APXS - Darker Green
- SS - Light Green
- IR Spec - Red
- Parachutes - White


IV) Payload Criteria

4.1 Selection, Design, and Verification of Payload Experiment

Science Payload:

Sample Methane-Ethane Lake (Experiment 1): The IR Spec works by exploiting the fact that molecules absorb frequencies that are characteristic of their structure. The absorption occurs at their resonant frequencies. The IR Spec also comes with the added benefit of being able to work with both liquid and solid samples. The FDNMR works by having an oscillating magnetic field to resonate with the nuclei of atoms. The FDNMR works at cryogenic temperatures and there is one being designed specifically for Titan, allowing for the FDNMR to function without the need for additional heating from the MMRTG.

Find Life (Experiment 2): FDNMR, SC, IR Spec, and Temperature Sensor. The FDNMR and IR Spec can perform the same tasks as mentioned in Experiment 1 to locate the presence of lipids and proteins which would be an indicator of the life currently or at a different point in time. The SC will likely be looking for signs of thermal vents and the change of temperature will be confirmed by the N-type Thermocouple. Based on Methane and Ethane being bad solvents, the mares will be clearer than the liquids on earth.

Map Mare Floor (Experiment 3): SS, SC. The SS will be the main instrument to map the mare floor, to be used as an active sonar to map distance from the submarine. The SC will be used to see and confirm that the data received by the SS.

Bed Composition (Experiment 4): BSA, IR spec, APXS. The BSA will consist of a series of linear actuators able to extend out about 2 ft in the form of a 4 degrees of freedom (DOF) arm with an extra joint to change the angle of the gripper, from there it will be able to pick up an object and send it to either the IR Spec or the APXS. The IR Spec works as stated in Experiment 1, the APXS works by irradiating the solid sample with alpha particles and X-rays from radioactive sources. The atoms have the ability to absorb the alpha particles to varying extents and is how the APXS is able to differentiate between molecules.

After the baseline of a 150 degree 10 second panorama is completed, experiments will be run based on priority.

Mobility Systems:

The Rover Compute Element (RCE) has been a staple of successful missions and will be responsible for controlling THANOS in this mission as well. 2 RCE's will go on the craft but only 1 will work at a time, as the other will remain dormant until something happens to the primary RCE. The primary RCE will serve the purpose of sending and receiving data from the sensors and antennas, controlling the prop motors and BSA actuators, and controlling the pumps to fill the ballast tanks and provide samples for the sensors.

THANOS will be powered by the MMRTG and 2 24v 10Ah, with the batteries in parallel, the craft will be able to run for a little over 2 hours assuming its using all of its equipment, then go to the surface to recharge its batteries and send data to the orbiter. On the SS and SC experiment to map the floor, the batteries can support mapping for 6 hours total powering the RCE's, motors, SS and SC. The APXS requires the most time to retrieve data. With highly detailed experiments going over 10 hours. This is the only instrument that would need to be run by both the Batteries and MMRTG in parallel. Giving the APXS a run time of 12 hours before running into energy deficiencies problems. Afterwards, the MMRTG will be able to recharge the batteries in 5 hours.

Recovery Systems:

Referencing the chart in section 1.3, The FDNMR and IR Spec can cover for one another if the other fails. Similarly the IR Spec and APXS can cover each other in a similar way. These 3 instruments can also verify what others are detecting. The dual RCE's can cover each other if one of them happens to fail. The 2 cameras on the ship can act as stereo vision to allow it to cover the SS in the event it fails. There will be 2 propellers on THANOS, that will be able to cover for one another as well.

4.2 Payload Concept Features and Definition of Purpose Critical to Mission

The unique features on THANOS will include 2 new technologies, a side scan sonar (SS) capable of operating at cryogenic temperatures, and the Force-Detected Nuclear Magnetic Resonance (FDNMR). The FDNMR on Earth requires cooling from liquid nitrogen and other cryogenic liquids; however, on Titan the FDNMR can just be run off of the passive cooling from the ambient temperature.

The instruments were chosen based off either their heritage or new technology,

Heritage instruments and equipment include: APXS, IR Spec, SC, UHF antenna, Low Gain antenna, Transceiver, MMRTG, and RCE

New instruments and equipment include: FDNMR, SS, and the BSA

The FDNMR is currently being researched by CalTech and JPL to work exclusively on the naturally cryogenic temperatures of Titan, the main challenge with the FDNMR, other than making it small and lightweight, is magnetic influence on the submarine from the FDNMR

The SS on earth is typically rated to go down to 0 degrees Celsius due to water becoming ice at that temperature, although some can go down to -40 degrees Celsius. This is nowhere near close enough to withstand the surface temperatures on Titan, let alone the sub-surface temperatures the craft will experience while submerged for extended periods of time.

The BSA is a 4 DOF arm composed of a series of linear actuators to be able to fold into the submarine, making it very different from the typical multi link design from the Mars rovers and ISS's Mobile Servicing System. Problems it may face on Titan other than size, weight, and cryogenic temperatures tolerance is the ability to fold neatly into the sub and the ability to pass the solid samples to either the APXS or the IR Spec while minimizing the noise it may create from its own composition.

4.3 Science Value

When THANOS lands on Titan, its first objective is to complete a 150 degree panorama, this operation will take about 10 seconds after landing. From there, experiments will be conducted based on priority. The table in section 3.3 has the list of the runtimes to complete a sampling from each instrument. After completing experiments 1 & 2 , THANOS will submerge itself, checking its height with the SS. Once it gets a clear rendering of the floor for Experiment 3, it will then continue to Experiment 4. If it reaches its depth limit before getting a clear rendering of the floor, THANOS will move towards the north shore until it gets a clear rendering for Experiment 3 and continue to Experiment 4.

Instruments used for each experiment by priority goes as follows:

Experiment 1: Liquid Composition. FDNMR, IR Spec

Experiment 2: Finding Life. FDNMR, SC, IR Spec, and N-type Temperature Sensor

Experiment 3: Mapping the Mare Floor. SS, SC

Experiment 4: Determining Bed Composition. BSA, APXS, IR Spec

4.4 Safety and Environment (Payload)

The safety officer will be lead by one of the engineering's team Omar Asbun.

The safety officer is responsible for safe operation during the mission, it has to follow specific requirements to ensure that there will be no inconveniences during the procedure of the mission.

There are a variety of guidelines that the safety officer has to follow to make sure that mission is clear to proceed. Which are as follows:

- Examine the basic model structure and recovery system. The primary safety hazard during this process are the electronic systems.

- Another safety is the payload, it's important that the payload does not separate under its own weight.
- Is the model stable? It's important to find the CG of the flight model which is its balance point
- Recovery system hardware, needs to be strong enough for recovery loads and mounted to solid structure. All fasteners should be inspected for tightness.
- Make sure that the parachute is sufficiently large to safely recover the model, also is the parachute protection from the ejection charge adequate and nonflammable
- Are receiver antennas protected from breakage
- Make sure if electronics are used for control functions in the model. If so, examine the electronics for items that may dislodge or break during flight.
- If timers initiate recovery system deployment, how is the delay time calculated.

Also due to the use of chemical devices there has to be a consideration of the chemical hazards. We have to make sure that the chemicals are correctly handled. Also due to the use of radioactive devices like the APXS and MMRTG that emit a low amount of radiation. For the radiation safety, we have to make sure that there are low-emitting radiation instruments and protective clothing

Event	Procedure
Failed/broken Propeller	Keep using the remaining propeller
2 Failed/Broken Propeller	Travel using currents at different heights
Failed FDNMR	Continue Experiment 1 with the IR Spec
Failed IR Spec	Continue Experiment 1 and 2 with the FDNMR and APXS respectively
Failed APXS	Continue Experiment 2 with IR Spec
Failed SS	Use both SC’s as stereovision to continue to map the floor
Failed bottom SC	Use SS to see what the BSA is picking up
Failed front SC	High angle of attack to use SS and SC bottom to see in front of THANOS

V) Activity Plan

5.1 Show status of activities and schedule

L'SPACE Budget								
Engineering And Science (Pre-Launch)	**2019**	**2020**	**2021**	**2022**	**Science (Post-Launch)**	**2031**	**2032**	
Year	**Yr 1 Total**	**Yr 2 Total**	**Yr 3 Total**	**Yr 4 Total**		**Yr 14 Total**	**Yr 15 Total**	**Cumulative Total**
PERSONNEL								
Scientist ($80,000/yr)	$320,000	$329,600	$339,488	$349,673		$360,163	$370,968	$2,069,891
Engineer ($80,000/yr)	$320,000	$329,600	$339,488	$349,673		$270,122	$278,226	$1,887,109
Total Salaries	$640,000	$659,200	$678,976	$699,345		$630,285	$649,193	$3,957,000
ERE (Fringe)								
Scientist	$89,312	$91,991	$94,751	$97,594		$100,521	$103,537	$577,707
Engineer	$89,312	$91,991	$94,751	$97,594		$75,391	$77,653	$526,692
Total ERE	$178,624	$183,983	$189,502	$195,187		$175,913	$181,190	$1,104,399

TOTAL PERSONNEL	$818,624	$843,183	$868,478	$894,533		$806,197	$830,383	$5,061,398
OTHER DIRECT COSTS								
Total Materials and Supplies	$27,000	$0	$0	$0		$0	$0	$27,000
Publications	$6,750	$0	$0	$0		$2,250	$2,250	$11,250
Total Travel	$25,416	$0	$0.00	$0		$0	$0.00	$19,608
Total Services	$0	$0	$0	$0		$0	$0	$0
Total Equipment	$125,709,722	$0.00	$0	$0.00		$0	$0.00	$125,709,722
Total Subcontracts (if you have contractors)	$160,000	$164,800	$169,744	$174,836		$0	$0	$669,380
Total Participant Support	$5,000	$5,150	$5,305	$5,464		$0	$0	$20,918
Tuition Remission (not applicable)	$0	$0	$0	$0		$0	$0	$0
Total Direct Costs	$126,	$1,01	$1,04	$1,07		$808,	$832,	$131,519,277

	752,512	3,133	3,527	4,833		447	633	
Total MTDC (see # 6 below)	$877,790	$843,183	$868,478	$894,533		$808,447	$832,633	$131,519,277
Total Subcontract F&A	$16,000	$16,480	$16,974	$17,484		$0	$0	$66,938
College or University F&A	$87,779	$84,318	$86,848	$89,453		$80,845	$83,263	$512,506
Total F&A	$103,779	$100,798	$103,822	$106,937		$80,845	$83,263	$579,444
Total Project Cost	$126,856,291	$1,113,931	$1,147,349	$1,181,769		$889,292	$915,897	$132,104,529
FED FLOW THROUGH (JPL, ARC, etc.)	$0	$0	$0	$0		$0	$0	$0
Overall Budget Margin (30%)					$39,545,837			$39,545,837
Total Project Cost	**$126,856,291**	**$1,113,931**	**$1,147,349**	**$1,181,769**	**$39,545,837**	**$889,292**	**$915,897**	**$171,650,366**

F&A % (see #4 below)	10%	10%	10%	10%		10%	10%	
Inflation Factor (How much costs will grow each year)	0%	3%	3%	3%		3%	3%	
Annual Wages								
Scientist Wages ($80,000 yr.)	80,000	82,400	84,872	87,418		90,041	92742	517,473
Engineer ($80,000 yr.)	80,000	82,400	84,872	87,418		90,041	92,742	517,473
<u>Level of Effort</u>								
Scientist (e.g., 1.0 = I FTE (Full-time Equivalent) across all years as an example)	4	4	4	4		4	4	
Engineer (e.g., 1.0 in yrs. 1,2 and .5 FTE in yrs. 3,4,5 and 0 FTE in yrs. 6)	4	4	4	4		3	3	
ERE (Fringe) - Staff Rates	0.28	0.28	0.28	0.28		0.28	0.28	

Travel Worksheet								
Travel Detail: ASU - Coco Beach, FL								
Duration (5 days, 4 nights)								
Air-fare (Roundtrip Est. From: XX & To: XX		500						
Per diem - Hotel ($121 per night x 4 nights)		484						
Per diem - Food = $71 (day 1 and 5 (travel days) are @ 75% of that amount)		320						
x 5 days (n-4 nights)		804						
Car rental (compact)		36						
x 5 days (weekly rate+ taxes)		180						
Airport Parking, tolls, gas for rental vehicle		150						
Total (Per Person)		2,118						

Number of persons		12	(6 Team Members, 3 Scientists, and 3 Engineers)					
Total		**25,416**						

Budget Narrative:

Publications cost:

-https://science.nasa.gov/researchers/sara/faqs#20

Scientific Instruments and Equipment:

Force Detected Nuclear Magnetic Resonance spectrometer…..$5M ($6.25M with 25% research)

-https://www.the-scientist.com/technology-profile/taking-it-higher-55417

Alpha particle X-ray spectrometer………………….…....……$30M

-Provided By L'SPACE

Infrared spectrometer……………………...….….…...……$15M

-Provided By L'SPACE

Side-Scan Sonar…………………………….…...…………$40,000 ($50,000 with 25% research)

-https://www.edgetech.com/wp-content/uploads/2018/06/4125-SAR-Brochure-012219-LR.pdf

Camera……………………………………….……………..$10M (x2)

-Provided By L'SPACE

N type thermocouple……………………………….…………$106.80

-https://www.omega.com/en-us/sensors-and-sensing-equipment/temperature/sensors/thermocouple-probes/bt-000-bt-090/p/BT-000-K-2-3-4-444-1

MMRTG………………………………………………….………$54M

-https://trello-attachments.s3.amazonaws.com/5ce63cb612c466620b80ac79/5cf9c89f9611db849df2f34d/45772ce8d223aadcade2c39ef2f92aee/Draft%2BDiscovery%2B19_released.pdf

Ultra High Frequency Antennae……………………………....$3,750

-https://www.endurosat.com/cubesat-store/all-cubesat-modules/uhf-antenna/?change-currency=1564297988345

Ultra High Frequency Transceiver…………………………....$4,375

-https://www.endurosat.com/cubesat-store/cubesat-communication-modules/uhf-transceiver-ii/

Rover Computer Element……………………………….....…$200,000 (x2)

-https://en.wikipedia.org/wiki/IBM_RAD6000

Low Gain Antenna…………………………………………….$1,130

-https://www.cdw.com/product/panorama-bracket-mount-antenna-antenna/5505331#PO

Batteries……………………………………………………….$180 (x2)

-https://www.amazon.com/Aegis-Battery-Rechargeable-High-energy-applications/dp/B07PJ5CNMM/ref=sr_1_7?keywords=LI-ion+24v&qid=1564212975&s=sporting-goods&sr=1-7

Machining Cost…………………………………………………$27,000

-https://www.custompartnet.com/estimate/machining/

Travel Costs:

Per Diem Rates:

-https://www.gsa.gov/travel/plan-book/per-diem-rates/per-diem-rates-lookup/?action=perdiems_report&state=FL&fiscal_year=2019&zip=&city=Cape%20Canaveral

Rental Car:

-https://book.carrentals.com/offers?dropoff_datetime=2019-10-17T10%3A00%3A00Z&dropoff_location=MLB&pickup_datetime=2019-10-13T10%3A00%3A00Z&pickup_location=MLB&sort=Price

DFW International Airport Rates:

-https://www.delta.com/cart/activity/tripsummary.action?cacheKeySuffix=4b41c61e-3ab1-4d5c-8402-60b3c2acd6d2&app=sl-sho&itinSegment[0]=0:E:DFW:ATL:DL:2255:OCT:13:2019:08A&mkcpgn=MS_GFSR_DESK_DOM_TR_DL_140414_GFSDESK_US&itinSegment[1]=0:E:ATL:MLB:DL:1325:OCT:13:2019:12P&paxCount=1&currencyCd=USD&itinSegment[2]=1:E:MLB:ATL:DL:1325:OCT:17:2019:02P&itinSegment[3]=1:E:ATL:DFW:DL:1810:OCT:17:2019:04P&numOfSegments=4&cabin=BE&exitCountry=US&vendorID=Google&vendorRedirectFlag=true&tr

ipType=roundTrip&fareBasis=UAVNA0BQ:UAVNA0BQ:TAVNA0BQ:TAVNA0BQ&price=369.00

Austin-Bergstrom International Airport Rates:

-https://www.aa.com/booking/passengers;jsessionid=263E1718CC4E5FD5872AFA858C5C7D7A?fromMetaSearch=true&bookingPathStateId=1560615094244-808&ischangedfare=false&ischangedservice=false&locale=en_US&c=MSE%7CGFS%7C20190615%7CDST%7CDST%7CWEB%7C%7CLLGG%7CLLGG%7CGFSE-0_0_0_1

Mission Concept Schedule:

The schedule for this mission concept is split into five phases within throughout the course of five years, of which include this preliminary design review, launch, data collection, and closeout.

Phase 1: Concept and Preliminary Design (2019)

In June of 2019, L'SPACE Team 19 was selected to carry out an in depth concept design that will help characterize a hydrocarbon sea, or mare, on Saturn's Moon Titan. During Phase 1, Team 19 will focus on developing the final concept design along with the necessary technology requirements and program plans. Once this has been completed and is allowed to move further, Team 19 will begin to transition into Phase 2 of this timeline.

Phase 2: Design Fabrication (2020)

In this phase, Team 19 will focus exclusively on the design reviews for our science instrumentation along with our engineering systems. Once this is completed, fabrication will begin for all of our flight hardware, software, and science instruments. Along with developing extensive procedures for the system assembly, every thing will be tested to ensure that they meet and exceed the design standards. From this, Team 19 will now shift into Phase 3 of the timeline.

Phase 3: System Assembly and Launch (2021 - 2022)

This phase emphasizes the importance of assembling all of the flight and science hardware so that the spacecraft is functioning properly. After extensive testing is completed to ensure that safety and design standards are met, we can then continue to launch. Through the span of a nine year window, Team 19's spacecraft will arrive to Saturn's moon Titan, where it will then begin data collection.

Phase 4: Data Collection and Sustainment (2031)
The spacecraft Team 19 has designed serves to act as a submarine, deploying at Titan's largest sea, Kraken Mare. Using the instrumentation on board the spacecraft, it will collect the data within the time span of ideally 30 minutes. Once the data is collected, it will then relay it back to Earth where it will be used for further analysis.

Phase 5: Conclusion (2032)
Once all of the necessary data is collected, processed, and archived, the spacecraft can be dismissed and Team 19's mission can finally come to an end. From this data, Team 19 hopes that its discoveries will be substantial and could one day contribute to further studies and exploration being done on Titan, a moon that has so much to contribute in the realm of space exploration.

Mission Concept Outreach:

Outreach for this mission concept will consist of general posters, flyers, posts on social media sites. On NASA's side, the animated part of the mission will be a key part of the outreach. With this outreach, a majority of the audience desired to be reached will be young adults. The emphasis on the use of social media and websites on this mission can be used to connect this mission to the younger generation, in ways to make them more interested in space exploration, and maybe even as far as inspiration for some to direct them into a career path. Presentations of this mission could be made to be given at several conferences on different topics and at colleges to expand the amount of interest and awareness in the field of space exploration.

VI) Conclusion

6.1 Summarize all aspects of your Mission Concept.

This mission is meant to withstand the harsh conditions that Kraken Mare presents. With the ability to dive underneath the surface our craft will be able to gather data that no other satellite has been able to because of the thick atmosphere. The objective of the mission is to answer 4 different questions that we currently have about Titan. The first objective is to sample the lake of Kraken Mare and determine its exact composition. The FDNMR and IR Spectrometer will be used to collect the data. The second objective is to find life in Kraken Mare using the FDNMR, SC, IR Spec, and Temperature Sensor to search for signs of life. The third objective is to map the bottom of the lake floor using the SS and SC. the fourth objective is to determine the composition of the lake bed using the BSA, IR spec, and APXS. These mission objectives were chosen because they each provide valuable data that otherwise could not be collected by current means.

The craft has gone through multiple iterations as the concept has progressed and the final design is a submarine that has been made to fit each instrument while staying within the required specifications. Once splashing down on the surface the initial startup procedures will commence and THANOS will complete its first objective of taking its panorama of the surface. Once complete the rest of the missions will be completed based on their priorities listed in the table in section 3.3.

The project timeline for the mission is from 2019 - 2032 or longer. Towards the end of phase five we will propose more missions for the craft to further the mission timeline and increase the science return on the investment of the spacecraft.

Modified from NASA’s Student Launch Initiative PDR Document -
https://www.nasa.gov/pdf/206051main_Preliminary_Design_Review_Req_508.pdf